森林火灾后植被绿度与树冠覆盖的变化差异及数学预测

Post-Fire Differences Between Vegetation Greenness and Tree Canopy Cover

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摘要

森林火灾后，植被重新变绿与树冠覆盖变化是否沿着相同轨迹？本研究使用美国 81 场火灾的固定受灾与对照位置，以灾前三年为基准，计算两项遥感指标：LG 为经过参照校准的绿色覆盖估计，ZP_TCC 为 USFS 树冠覆盖比例。DE 为树冠覆盖相对状态减 LG 相对状态，展示时乘 100，以“点”计。

36 场火灾用于拟合二次函数，37 场用于回顾性检验，8 场用于补充观测核对。研究比较二次函数与利用火后第一年实际观测的幅度上限校正，预测窗口为火后 1—8 年，误差评价限第 2—8 年。

首年 ZP 在 46—64 点外时完整平移 ZP，区间内保留四分之一平移。仅在 ZP 区间外且 LG 首年绝对偏差至少 12 点时，保留 LG 偏差方向，并将平移幅度限制为不超过 ZP；其他 LG 采用衰减校正。37 场检验中，LG、ZP、DE 的 RMSE 分别为 8.56、8.42、8.60 点，R^2分别约 0.616、0.621、0.262。

训练交叉验证中，校正降低了 LG 和 ZP 误差，但 DE RMSE 由二次函数的 8.71 升至 8.88 点；回顾性检验中 DE 则由 9.59 降至 8.60 点。不同评估样本的表现不一致，说明校正的普遍预测优势尚未得到充分证明。

Park、Dixie、Caldor、Creek 的实际首年输入和后续试算均提供。预测限火后 1—8 年，评分仅含第 2—8 年，不代表完整森林生态恢复。

关键词　森林火灾　植被绿度　树冠覆盖　首年校正　预测性能

1 引言

草本、灌木、萌蘖和幸存树冠都能贡献绿色，而绿色覆盖则不能代表森林本身的生态恢复情况。而本研究选取树冠覆盖作为简化的代表生态恢复的指标。比较两项相对状态，可以研究绿色重新出现与树冠变化之间的差异。平均函数提供参照；首年实际观测则可能帮助识别某场火灾偏离平均曲线的程度。

本研究检验首年观测能否帮助预测随后七年的遥感相对状态，分别报告训练验证和回顾性检验结果。地区、气候和更新方式可能影响轨迹，但本文不将函数形状解释为生态恢复机制。

Post Fire Differences Between Vegetation Greenness and Tree Canopy Cover

Temporal patterns and mathematical prediction

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Abstract

Do vegetation regreening and changes in tree canopy cover follow the same trajectory after forest fires? This study uses fixed burned and control locations for 81 fires in the United States, with the three years before each fire as the baseline. Two remotely sensed indicators are calculated: LG is a reference-calibrated estimate of green cover, and ZP_TCC is the USFS tree canopy cover fraction. DE is the relative state of canopy cover minus the relative state of LG. Displayed values are multiplied by 100 and expressed in points.

Quadratic functions are fitted using 36 fires; 37 fires are used for retrospective testing and eight for supplementary checks of available observations. The study compares quadratic functions with an amplitude cap correction that uses actual observations from the first year after each fire. Predictions cover post-fire years 1–8, while errors are evaluated only for years 2–8.

When first-year ZP_TCC is outside 46–64 points, its curve receives the full first-year offset; within that interval, it receives one quarter of the offset. Only when ZP_TCC is outside the interval and the absolute first-year LG error is at least 12 points is the LG offset capped at the magnitude of the ZP_TCC shift, retaining the direction of the LG error. Other LG corrections decay with time. Across the 37 test fires, RMSE is 8.56, 8.42 and 8.60 points for LG, ZP_TCC and DE, respectively; the corresponding R^2 values are approximately 0.616, 0.621 and 0.262.

In training cross-validation, the correction reduces LG and ZP_TCC errors, but DE RMSE increases from 8.71 points for the quadratic function to 8.88 points. In retrospective testing, DE RMSE instead decreases from 9.59 to 8.60 points. This difference between evaluation samples means that a general predictive advantage has not been established.

Actual first-year inputs and subsequent predictions are provided for Park, Dixie, Caldor and Creek. Predictions are restricted to post-fire years 1–8, and scoring to years 2–8. They do not represent complete ecological recovery of a forest.

Keywords Forest fire Vegetation greenness Tree canopy cover First-year correction Predictive performance

1 Introduction

Herbs, shrubs, resprouts and surviving tree crowns can all contribute green cover, so green cover alone cannot represent the ecological recovery of a forest. This study uses tree canopy cover as a simplified indicator of recovery. Comparing the two relative states allows differences between regreening and canopy change to be examined. An average function provides a reference, while actual first-year observations may help identify how far an individual fire departs from that average curve.

The study tests whether first-year observations help predict remotely sensed relative states over the following seven years. Training validation and retrospective test results are reported separately. Region, climate and regeneration strategy may affect trajectories, but the shape of a fitted function is not interpreted as evidence of an ecological recovery mechanism.

2 指标与相对状态

2.1 理论结构目标与实际代理

理论物理 ZP 为单位面积内胸径至少 10 厘米的活立木断面积密度，单位平方米／公顷：

$$ZP_{\text{phys}} = \frac{10000}{A}\sum_{i=1}^{n}\pi\left(\frac{DBH_i}{200}\right)^2 \quad (\text{m}^2/\text{ha})$$

它可通过地面逐树调查直接测量，但目前不能为全部研究火灾建立经充分验证的数据库。因此时间模型采用 ZP_TCC＝TCC／100；它是树冠覆盖比例，不代表活树数量、树高或断面积。[1]

2.2 绿色覆盖估计

LG 由 DIRECT 绿色覆盖参照确定，N、E 分别为 MODIS NDVI、EVI：[2—3]

$$LG = \text{clip}_{[0,\,1]}(-0.149100557 + 0.554620599N + 0.727351848E)$$

LG 计算公式的系数来自绿色覆盖参照校准，时间函数拟合不调整这些系数。clip 将原始绿色覆盖估计限制在 0—1；LG 存在航空迁移偏差，并非所有火灾的地面真值。NDVI、EVI 和胸径等测量量详见附录 A。

2.3 灾前三年基准与 DE

I_k(t)、C_k(t)分别为固定受灾、对照位置的年度平均。先计算两组比值 q，再除以火前三年 q 的算术平均 b：

$$q_k(t) = I_k(t)/C_k(t), \qquad b_k = \frac{1}{3}\sum_{u=-3}^{-1} q_k(u)$$

$$R_k(t) = q_k(t)/b_k, \qquad DE(t) = R_{ZP_{\text{TCC}}}(t) - R_{LG}(t)$$

LG 和树冠覆盖各用自己的基准。R＝1 只表示回到灾前相对基准，不能证明全部生态恢复；R 可以超过 1，未截断。DE 小于 0 表示 LG 相对状态更高，DE 大于 0 则表示树冠覆盖相对状态更高。

文中展示值均乘 100，以“点”计。46 点和 64 点指 R_ZP_TCC 的展示数值，不是原始 TCC 为 46%或 64%；12 点指 LG 相对状态预测偏差的绝对值。DE 不是两种恢复速度的直接差值。

2 Indicators and relative states

2.1 The physical structural target and its practical proxy

The theoretical physical indicator ZP is the basal area density of living trees with diameter at breast height of at least 10 cm, measured in square metres per hectare:

$$ZP_{\text{phys}} = \frac{10000}{A}\sum_{i=1}^{n}\pi\left(\frac{DBH_i}{200}\right)^2 \quad (\text{m}^2/\text{ha})$$

It can be measured directly through field inventories of individual trees. However, a sufficiently validated database is not currently available for all study fires. The time model therefore uses ZP_TCC = TCC/100, the tree canopy cover fraction. It does not represent live-tree counts, tree height or basal area. [1]

2.2 Estimating green cover

LG is calibrated against DIRECT green-cover references. N and E denote MODIS NDVI and EVI, respectively: [2–3]

$$LG = \text{clip}_{[0,\,1]}(-0.149100557 + 0.554620599N + 0.727351848E)$$

The coefficients in the LG formula come from calibration against green-cover references and are not adjusted when the time functions are fitted. The clip operation bounds the raw green-cover estimate between 0 and 1. LG shows bias when transferred to airborne reference data and is not ground truth for every fire. NDVI, EVI, stem diameter and the other measurement quantities are explained in Appendix A.

2.3 The three-year pre-fire baseline and DE

I_k(t) and C_k(t) are the annual means at fixed burned and control locations, respectively. Their ratio q is calculated first, then divided by b, the arithmetic mean of q over the three pre-fire years:

$$q_k(t) = I_k(t)/C_k(t), \qquad b_k = \frac{1}{3}\sum_{u=-3}^{-1} q_k(u)$$

$$R_k(t) = q_k(t)/b_k, \qquad DE(t) = R_{ZP_{\text{TCC}}}(t) - R_{LG}(t)$$

LG and canopy cover each use their own baseline. R = 1 means only that the indicator has returned to its relative pre-fire reference; it does not demonstrate complete ecological recovery. R may exceed 1 and is not truncated. Negative DE means that the relative state of LG is higher; positive DE means that the relative state of canopy cover is higher.

All displayed values are multiplied by 100 and expressed in points. The thresholds of 46 and 64 points refer to displayed R_ZP_TCC values, not raw TCC values of 46% or 64%. The 12-point threshold refers to the absolute prediction error in the first-year relative state of LG. DE is not a direct difference between two recovery rates.

3 数据与研究范围

3.1 数据来源与固定空间位置

研究资料包括 81 场美国森林火灾、636 个火后年度状态和 9,564 个固定位置。36 场火灾用于函数拟合，37 场用于回顾性检验，其余 8 场用于补充观测核对。[4—6]

MOD13Q1.061 提供 250 米、16 天 NDVI 与 EVI。选取每年 6—8 月 SummaryQA 为 0 或 1 的记录，中位数合成为年度夏季指数。USFS TCC v2025-6 的 Science 树冠覆盖产品为 30 米；data_mask＝1 的有效覆盖先平均到 250 米网格，再平均固定位置。[1—2]

火前三年与火后各年都使用同一批位置。保持缺测，不用预测补观测；对照扰动、低基准、不稳定基准和产品年份均保留核查记录。官方 TCC 该版本覆盖 1985—2025 年，因此 2026 年的 LG 可以已有观测而 TCC 仍然缺少。[1]

3.2 训练与检验样本

36 场拟合火灾均具有完整八年记录，DE 呈下降轨迹。37 场回顾性检验和 8 场补充记录的用途见表 1。第一年是校正输入，因此预测误差从第 2 年开始计算。

样本用途	火灾数	观测与评价
函数拟合及交叉验证	36	完整八年记录
回顾性检验	37	第 2—8 年评价
补充观测	8	按已有年份核对

表 1 研究样本的用途。77 场具有完整八年观测。Carr、Creek、Dixie、Caldor 的观测期尚不足八年，已有观测用于补充核对，未来年份只给出预测。

训练交叉验证、37 场回顾性检验和补充观测的结果分别报告。RMSE、MAE、偏差、SD、R^2 和预测区间覆盖率均注明对应样本与时间窗口。

3 Data and study scope

3.1 Data sources and fixed sampling locations

The study dataset includes 81 forest fires in the United States, 636 post-fire annual states and 9,564 fixed locations. Of these fires, 36 are used to fit the functions, 37 for retrospective testing and the remaining eight for supplementary observation checks. [4–6]

MOD13Q1.061 provides NDVI and EVI at 250 m resolution in 16-day composites. Records from June–August with SummaryQA of 0 or 1 are selected; their medians form the annual summer indices. The USFS Science TCC product, version 2025-6, has 30 m resolution. Valid cover values with data_mask = 1 are first averaged onto the 250 m grid and then across the fixed locations. [1–2]

The same locations are used for the three pre-fire years and every post-fire year. Missing observations remain missing and are not replaced by predictions. Checks of control-area disturbance, low or unstable baselines, and product years are retained. This official TCC version covers 1985–2025, so LG observations may be available for 2026 while TCC is still missing. [1]

3.2 Training and test samples

All 36 fitting fires have complete eight-year records and declining DE trajectories. The uses of the 37 retrospective test fires and eight supplementary records are shown in Table 1. Because the first year is a correction input, prediction errors are calculated from year 2 onward.

Sample use	Fires	Observations and evaluation
Fitting and cross-validation	36	Complete eight-year records
Retrospective test	37	Years 2–8 evaluated
Supplementary observations	8	Available years checked

Table 1. Uses of the study samples. Complete eight-year observations are available for 77 fires. Carr, Creek, Dixie and Caldor have not yet reached eight years of observation. Their available observations are used for supplementary checks, and future years contain predictions only.

Training cross-validation, the 37-fire retrospective test and supplementary observation checks are reported separately. RMSE, MAE, bias, SD, R^2 and prediction-interval coverage are identified with their corresponding sample and time window.

4 二次函数与首年信息

4.1 训练与验证

二次函数在 36 场训练火灾内拟合，每场八年记录的总权重相同。训练验证以起火年份分组，每次留出同一年起火的全部训练火灾，用其他年份拟合后预测留出的火灾。[10]

LG 与 ZP_TCC 分别拟合二次函数，DE 由两个预测分量相减得到。二次函数表示训练样本的平均轨迹，适用窗口为火后 1—8 年。

$$f_{LG}(t) = 0.457185239 + 0.100472584t - 0.006583693t^2$$

$$f_{ZP_{\mathrm{TCC}}}(t) = 0.549880554 + 0.016052486t - 0.001234201t^2$$

$$\widehat{DE}(t) = 0.092695314 - 0.084420099t + 0.005349492t^2$$

以上三式为二次函数及其差值，输出均为相对状态。利用第一年观测后，按下述规则加入校正量，得到各场火灾的预测函数。

4.2 已知第一年之后的预测

令 e_k 为该场第一年实际相对状态与函数第一年预测的差值：

$$e_k = R_k(1) - f_k(1)$$

$$\widehat{R}_k(t) = f_k(t) + s_k(t), \qquad t = 2, \ldots, 8$$

校正使用 t＝1 的实际观测，但只评价 t＝2—8。幅度上限不强制 LG 函数通过首年观测点；第一年仍不计入预测精度。相对状态和预测范围均不截在 0—100 点。

首年信息可能包括真实初始差异，也可能包含云影、产品误差和对照变化。因此校正是一项数学假设，需要验证其在后续年份是否仍有用；不能自动当作生态机制。

4 Quadratic functions and first-year information

4.1 Fitting and validation

The quadratic functions are fitted to the 36 training fires, with equal total weight for each fire's eight-year record. Training validation is grouped by ignition year: all training fires that began in the same year are held out together, and functions fitted to the other years predict those held-out fires. [10]

Separate quadratic functions are fitted for LG and ZP_TCC; predicted DE is obtained by subtracting the two components. The quadratic functions describe the average trajectory of the training sample and apply only to post-fire years 1–8.

$$f_{LG}(t) = 0.457185239 + 0.100472584t - 0.006583693t^2$$

$$f_{ZP_{\mathrm{TCC}}}(t) = 0.549880554 + 0.016052486t - 0.001234201t^2$$

$$\widehat{DE}(t) = 0.092695314 - 0.084420099t + 0.005349492t^2$$

The three expressions above are the two quadratic functions and their difference; all outputs are relative states. First-year observations are then used to add corrections according to the rules below, producing a prediction function for each fire.

4.2 Prediction after observing the first year

Let e_k be the difference between a fire's actual first-year relative state and the function's first-year prediction:

$$e_k = R_k(1) - f_k(1)$$

$$\widehat{R}_k(t) = f_k(t) + s_k(t), \qquad t = 2, \ldots, 8$$

The correction uses the actual observation at t = 1, but evaluation includes only t = 2–8. The amplitude cap does not force the LG curve through the first-year observation, and the first year is excluded from predictive accuracy. Relative states and prediction ranges are not truncated to 0–100 points.

First-year information may contain real initial differences as well as cloud shadows, product errors and changes in the controls. The correction is therefore a mathematical hypothesis whose usefulness in later years must be tested; it is not automatically an ecological mechanism.

4.3 首年校正的分段规则

令 s_ZP、s_LG 为加到两项二次函数上的校正量。首年 ZP 在[0.46,0.64]之外时，s_ZP＝e_ZP；区间内时 s_ZP＝0.25e_ZP。LG 按表 2 计算：

对象	条件成立时的校正量	其他情况
ZP	R_ZP(1)＜0.46 或＞0.64： s_ZP＝e_ZP	s_ZP＝0.25e_ZP
LG	ZP 区间外，且|e_LG|≥0.12： s_LG＝sign(e_LG)×min(|e_LG|,|s_ZP|)	s_LG＝e_LG×0.5^(t−1)

表 2 幅度上限校正规则。条件成立时，s_LG＝sign(e_LG)×min(|e_LG|,|s_ZP|)；sign 保留 LG 偏差的正负方向，min 取两个绝对值中较小者。其他 LG 使用衰减校正。常数平移只改变截距；DE 始终由两项预测相减得到。

4.3 Piecewise first-year correction rules

Let s_ZP and s_LG be the corrections added to the two quadratic functions. If first-year ZP_TCC lies outside [0.46, 0.64], s_ZP = e_ZP; within that interval, s_ZP = 0.25e_ZP. LG is corrected as shown in Table 2.

Indicator	Correction when the condition is met	Otherwise
ZP	R_ZP(1) < 0.46 or > 0.64: s_ZP = e_ZP	s_ZP = 0.25e_ZP
LG	ZP outside the interval and |e_LG| ≥ 0.12: s_LG = sign(e_LG) × min(|e_LG|, |s_ZP|)	s_LG = e_LG × 0.5^(t−1)

Table 2. Amplitude cap correction. When the stated condition is met, s_LG = sign(e_LG) × min(|e_LG|, |s_ZP|). The sign function preserves the direction of the LG error; min takes the smaller absolute magnitude. Otherwise, the LG correction decays. A constant vertical shift changes only the intercept. DE is always obtained by subtracting the component predictions.

5 观测轨迹与预测误差

5.1 火灾后观测轨迹

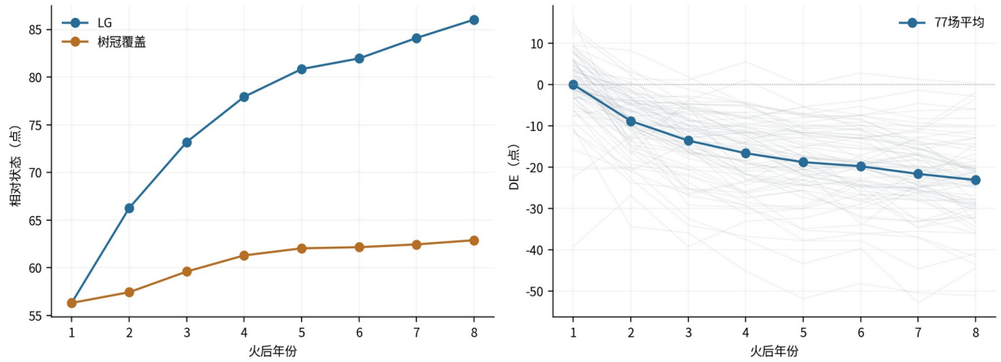


图 1 77 场具有完整八年观测的火灾。左图为 LG 与树冠覆盖的平均相对状态，右图为各场 DE 及平均 DE。每场每年贡献一个事件平均值，单位为点。

火后年份	平均 LG	平均树冠覆盖	平均 DE
1	56.33	56.31	-0.01
8	86.02	62.89	-23.13

这些均值使用相同 77 场完整八年案例，避免每年样本数不同影响平均轨迹。火灾之间仍存在明显差异，有些 DE 下降，有些基本不变或增加；平均曲线不能代表每一场火灾。

绿色植被恢复与树冠覆盖可能受不同过程影响，但覆盖差异本身不能判定幼树更新、树木死亡、采伐或气候的具体贡献。

5 Observed trajectories and prediction errors

5.1 Observed post-fire trajectories

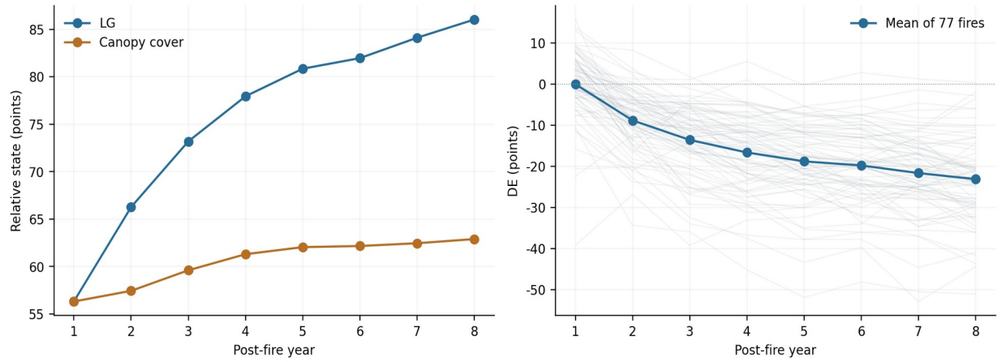


Figure 1. The 77 fires with complete eight-year observations. The left panel shows mean relative states of LG and canopy cover; the right panel shows each fire's DE and mean DE. Each fire contributes one event mean per year. Values are in points.

Post-fire year	Mean LG	Mean canopy cover	Mean DE
1	56.33	56.31	-0.01
8	86.02	62.89	-23.13

These means use the same 77 fires with complete eight-year records, avoiding changes in annual sample size that could distort the mean trajectory. Differences between fires remain substantial: DE declines in some fires, stays approximately unchanged in others and increases in others. The average curve cannot represent every fire.

Green vegetation recovery and canopy cover may be influenced by different processes. However, the cover difference alone cannot identify the specific contributions of seedling recruitment, tree mortality, logging or climate.

5.2 训练交叉验证

方案	LG RMSE	树冠覆盖 RMSE	DE RMSE
二次函数	10.67	11.29	8.71
幅度上限校正	9.26	7.08	8.88

表 3 36 场训练火灾、第 2—8 年的按起火年份交叉验证，单位为点。校正使 LG RMSE 由 10.67 降至 9.26 点，树冠覆盖由 11.29 降至 7.08 点；DE 由 8.71 升至 8.88 点，未显示改善。

5.3 回顾性检验

方案	LG RMSE	树冠覆盖 RMSE	DE RMSE
二次函数	12.66	13.65	9.59
幅度上限校正	8.56	8.42	8.60

表 4 37 场检验火灾、第 2—8 年的 RMSE，单位为点。第一年用作校正输入，不计入预测误差。

幅度上限校正使检验 LG RMSE 由 12.66 降至 8.56 点，树冠覆盖由 13.65 降至 8.42 点，DE 由 9.59 降至 8.60 点。

本研究为回顾性分析，研究资料也用于确定校正规则，因此这些结果不能视为严格前瞻盲测。训练验证与回顾性检验的 DE 表现不一致，模型对新火灾的普遍适用性仍需检验。

5.2 Training cross-validation

Method	LG RMSE	Canopy-cover RMSE	DE RMSE
Quadratic function	10.67	11.29	8.71
Amplitude cap correction	9.26	7.08	8.88

Table 3. Cross-validation by ignition year for the 36 training fires, evaluated over years 2–8, in points. The correction reduces LG RMSE from 10.67 to 9.26 points and canopy-cover RMSE from 11.29 to 7.08 points. DE RMSE rises from 8.71 to 8.88 points, showing no improvement.

5.3 Retrospective testing

Method	LG RMSE	Canopy-cover RMSE	DE RMSE
Quadratic function	12.66	13.65	9.59
Amplitude cap correction	8.56	8.42	8.60

Table 4. RMSE over years 2–8 for the 37 test fires, in points. The first year is used as a correction input and is excluded from prediction errors.

The amplitude cap correction reduces test RMSE from 12.66 to 8.56 points for LG, from 13.65 to 8.42 points for canopy cover, and from 9.59 to 8.60 points for DE.

This is a retrospective analysis, and study information also contributed to determining the correction rules. These results therefore do not constitute a strictly prospective blind test. The inconsistent DE performance between training validation and retrospective testing means that general applicability to new fires remains to be tested.

5.4 RMSE 与 R^2

目标	方案	RMSE	MAE	偏差	R^2
LG	二次函数	12.66	9.05	-1.48	0.160
树冠覆盖	二次函数	13.65	10.42	-1.38	0.005
DE	二次函数	9.59	7.27	0.10	0.081
LG	幅度上限校正	8.56	6.88	-1.66	0.616
树冠覆盖	幅度上限校正	8.42	6.12	-2.57	0.621
DE	幅度上限校正	8.60	6.95	-0.91	0.262

表 5 第 2—8 年，偏差＝预测－观测；RMSE、MAE、偏差以点计，R^2 无单位。相同观测的 SD 分别为 LG 13.82、ZP 13.68、DE 10.00 点。

$$R^2_{\text{test}} = 1 - \frac{\text{RMSE}^2}{\text{SD}^2}$$

本文 SD 用与 RMSE 相同的火灾等权、总体描述方差定义。因此图中近似式在此成为等式；不能换用不同样本或分母 N−1 的标准差而不作调整。

R^2＝1 表示无预测误差，0 表示与使用这批实际观测平均值具有相同平方误差，负值表示更差。该平均值是事后统计参照，并非预测时已知的数值。

幅度上限校正的 LG、ZP、DE R^2 约为 0.616、0.621、0.262。DE 的平方误差低于使用这批实际观测平均值的事后参照，但 R^2 不是命中率，也不代表生态机制的解释比例。

按起火年份整组重抽样，校正相对二次函数的树冠覆盖 RMSE 差为-5.23 点，95%范围为-8.96 至-1.50 点；DE 差为-1.00 点，范围为-3.76 至 0.64 点。这些范围表达本批样本的不确定性。

5.4 RMSE and R^2

Target	Method	RMSE	MAE	Bias	R^2
LG	Quadratic function	12.66	9.05	-1.48	0.160
Canopy cover	Quadratic function	13.65	10.42	-1.38	0.005
DE	Quadratic function	9.59	7.27	0.10	0.081
LG	Amplitude cap correction	8.56	6.88	-1.66	0.616
Canopy cover	Amplitude cap correction	8.42	6.12	-2.57	0.621
DE	Amplitude cap correction	8.60	6.95	-0.91	0.262

Table 5. Results for years 2–8. Bias = prediction − observation. RMSE, MAE and bias are in points; R^2 is dimensionless. For the same observations, SD is 13.82 points for LG, 13.68 for ZP_TCC and 10.00 for DE.

$$R^2_{\text{test}} = 1 - \frac{\text{RMSE}^2}{\text{SD}^2}$$

Here, SD is defined from the descriptive population variance using the same equal-fire weights as RMSE. The relationship shown is therefore an equality under this definition. An SD based on a different sample or an N−1 denominator cannot be substituted without adjustment.

R^2 = 1 indicates no prediction error. A value of 0 indicates the same squared error as using the mean of this set of actual observations; negative values indicate worse performance. That observed mean is a retrospective statistical reference, not a value known when a forecast is made.

R^2 for the amplitude cap correction is approximately 0.616 for LG, 0.621 for ZP_TCC and 0.262 for DE. The DE squared error is lower than the retrospective reference based on the mean of these observations. However, R^2 is neither a hit rate nor a proportion of ecological mechanisms explained.

Resampling whole ignition-year groups gives a canopy-cover RMSE difference of −5.23 points for the correction relative to the quadratic function, with a 95% range of −8.96 to −1.50 points. For DE, the difference is −1.00 point, with a range of −3.76 to 0.64 points. These ranges describe uncertainty in this sample.

5.5 每个年份的预测准确度

表 6 使用相同 37 场火灾。误差和 SD 以点计，R^2 按照本年观测 SD 计算。第一年是输入，不计入预测精度。

火后年份	二次 RMSE	上限校正 RMSE	观测标准差	上限校正 R^2
2	14.28	7.80	13.88	0.684
3	14.16	8.47	13.77	0.621
4	14.27	9.83	13.97	0.505
5	12.54	9.13	12.50	0.467
6	11.58	8.24	11.54	0.490
7	10.81	7.86	10.79	0.469
8	10.28	8.41	10.19	0.319

表 6a LG 逐年误差与 R^2。

火后年份	二次 RMSE	上限校正 RMSE	观测标准差	上限校正 R^2
2	13.97	4.23	13.91	0.908
3	13.75	6.43	13.74	0.781
4	13.36	8.11	13.24	0.625
5	13.17	8.75	13.03	0.549
6	13.26	9.27	13.13	0.502
7	13.75	9.80	13.58	0.479
8	14.27	10.63	13.99	0.423

表 6b 树冠覆盖逐年误差与 R^2。逐年 SD 只包括同龄火灾间差异；总体 SD 还包括七年间的变化，因此不能用总体 R^2 代替每一年的判断。

5.5 Predictive accuracy by year

Table 6 uses the same 37 fires. Errors and SD are in points; R^2 is calculated using that year's observed SD. The first year is an input and is excluded from predictive accuracy.

Post-fire year	Quadratic RMSE	Corrected RMSE	Observed SD	Corrected R^2
2	14.28	7.80	13.88	0.684
3	14.16	8.47	13.77	0.621
4	14.27	9.83	13.97	0.505
5	12.54	9.13	12.50	0.467
6	11.58	8.24	11.54	0.490
7	10.81	7.86	10.79	0.469
8	10.28	8.41	10.19	0.319

Table 6a. Annual LG errors and R^2.

Post-fire year	Quadratic RMSE	Corrected RMSE	Observed SD	Corrected R^2
2	13.97	4.23	13.91	0.908
3	13.75	6.43	13.74	0.781
4	13.36	8.11	13.24	0.625
5	13.17	8.75	13.03	0.549
6	13.26	9.27	13.13	0.502
7	13.75	9.80	13.58	0.479
8	14.27	10.63	13.99	0.423

Table 6b. Annual canopy-cover errors and R^2. Annual SD includes only differences among fires at the same post-fire age. Overall SD also includes change across the seven years, so overall R^2 cannot replace evaluation of each year.

5.6 DE 逐年误差与差值的作用

火后年份	二次 RMSE	上限校正 RMSE	观测标准差	上限校正 R^2
2	8.48	7.02	7.07	0.015
3	8.94	7.93	8.47	0.123
4	9.54	8.34	9.47	0.225
5	9.90	9.03	9.86	0.160
6	9.75	8.70	9.35	0.133
7	9.92	9.25	9.49	0.052
8	10.48	9.63	10.38	0.139

表 6c DE 逐年性能。某一年表现好，不代表整个窗口或所有火灾都预测准确。

DE 预测误差等于 ZP 误差减 LG 误差。两个校正量相差过大时，可能额外放大差值误差。限制 LG 平移幅度能够减小其中一部分影响，但不意味着两项真实状态必须相同，也不保证每场 DE 都改善。

DE 的规律仍值得研究，但当前差值不能自动消除地区、气候、树种或测量误差，也不能解释为恢复速度的直接差值。

5.7 经验预测范围

80%、90%、95%经验范围使用 36 场训练火灾按起火年份留出的校正预测绝对误差计算。将误差排序，取第 ceil[(36＋1)×目标覆盖率]个值为半宽；ceil 表示向上取整。同一指标、年份和目标对所有案例使用同一半宽。

半宽由训练误差计算，实际覆盖率由 37 场回顾性检验计算。训练只有 36 场，95%目标对应训练留出误差的最大值，个别年份较宽。这些经验范围不保证新事件达到标注覆盖率，第一年不画范围。

独立图集展示 81 场火灾的二次函数、幅度上限校正和实际观测。图中列明各场火灾的样本用途。

5.6 Annual DE errors and the role of the difference

Post-fire year	Quadratic RMSE	Corrected RMSE	Observed SD	Corrected R^2
2	8.48	7.02	7.07	0.015
3	8.94	7.93	8.47	0.123
4	9.54	8.34	9.47	0.225
5	9.90	9.03	9.86	0.160
6	9.75	8.70	9.35	0.133
7	9.92	9.25	9.49	0.052
8	10.48	9.63	10.38	0.139

Table 6c. Annual DE performance. Good performance in one year does not imply accurate prediction throughout the window or for every fire.

The DE prediction error equals the ZP_TCC error minus the LG error. Large differences between the two corrections can further amplify the error in their difference. Capping the LG shift can reduce part of this effect, but does not imply that the two actual states must be identical or guarantee improvement in DE for every fire.

Patterns in DE remain worth studying. However, the current difference does not automatically remove effects of region, climate, tree species or measurement error, and it cannot be interpreted as a direct difference in recovery rates.

5.7 Empirical prediction ranges

The 80%, 90% and 95% empirical ranges use absolute corrected-prediction errors from the 36 training fires held out by ignition year. After sorting the errors, the ceil[(36 + 1) × target coverage]th value is used as the half-width; ceil means rounding upward. Every case uses the same half-width for a given indicator, year and target coverage.

Half-widths are calculated from training errors, while actual coverage is calculated from the 37-fire retrospective test. With only 36 training fires, the 95% target uses the largest held-out training error, producing wide ranges in some years. These empirical ranges do not guarantee the stated coverage for new events. No range is drawn for the first year.

A separate atlas shows the quadratic functions, amplitude cap corrections and actual observations for all 81 fires. Each figure identifies how that fire is used in the study.

6 近期火灾的固定函数试算

6.1 可用的首年记录

Park 2024 使用 120 个固定位置的 NDVI、EVI 和 TCC 记录，按同一 LG 公式、TCC 比例、受灾与对照比值及灾前三年基准计算。Creek 2020、Dixie 和 Caldor 2021 使用相同指标与预测规则。[11—14]

火灾及首年	LG 首年	ZP 首年	校正分支
CREEK 2021	46.69	54.07	LG 衰减、ZP 四分之一
CALDOR 2022	43.48	53.37	LG 衰减、ZP 四分之一
DIXIE 2022	37.04	43.32	LG 幅度上限、ZP 完整
PARK 2025	27.76	31.65	LG 幅度上限、ZP 完整

表 7 首年输入为相对状态点数。Dixie 与 Park 同时满足 ZP 区间外和 LG 偏差至少 12 点，LG 使用幅度上限；Creek、Caldor 的 ZP 在区间内，LG 不触发该分支。

这四场不属于 36 场拟合样本或 37 场主检验样本。Creek、Dixie、Caldor 属于数据库的补充记录，Park 为近期应用案例。因来源和观测长度不同，其已有后续观测另列核对。

Gifford、Green 2025 已有 2026 年 LG 观测，但同一 TCC 产品版本没有 2026 年首年树冠覆盖记录，尚不能计算本方法所需的完整校正输入。

6 Applying the fixed functions to recent fires

6.1 Available first-year records

Park 2024 uses NDVI, EVI and TCC records from 120 fixed locations, with the same LG formula, TCC fraction, burned-to-control ratios and three-year pre-fire baseline. Creek 2020, and Dixie and Caldor 2021, use the same indicators and prediction rules. [11–14]

Fire and first observed year	First-year LG	First-year ZP_TCC	Correction applied
CREEK 2021	46.69	54.07	LG decay; one-quarter ZP shift
CALDOR 2022	43.48	53.37	LG decay; one-quarter ZP shift
DIXIE 2022	37.04	43.32	LG cap; full ZP shift
PARK 2025	27.76	31.65	LG cap; full ZP shift

Table 7. First-year inputs are relative states in points. Dixie and Park satisfy both conditions: ZP_TCC is outside the interval and the absolute LG error is at least 12 points. Their LG correction is capped. Creek and Caldor have ZP_TCC within the interval, so this LG branch is not triggered.

These four fires belong to neither the 36-fire fitting sample nor the 37-fire main test sample. Creek, Dixie and Caldor are supplementary database records; Park is a recent application case. Because their sources and observation lengths differ, available subsequent observations are checked separately.

Gifford and Green 2025 have LG observations for 2026. However, the same TCC product version has no first-year canopy-cover record for 2026, so the complete correction inputs required by this method are not yet available.

6.2 Park 与 Dixie 的走势

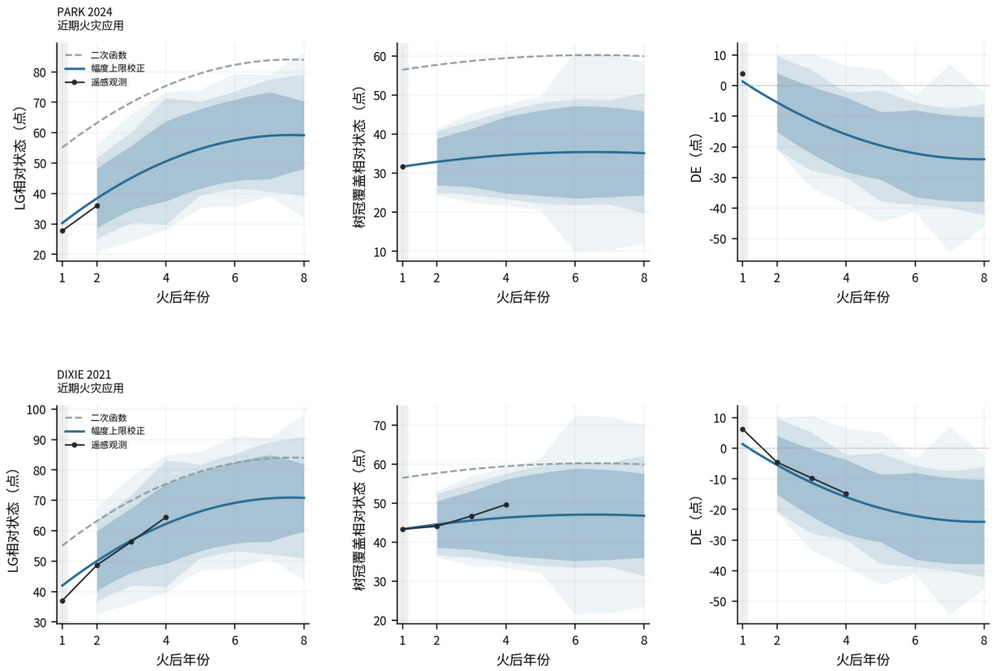


图 2 灰虚线为二次函数，蓝线为幅度上限校正，黑点为实际遥感观测。

Park 与 Dixie 的 ZP 完整平移，LG 保留偏差方向，平移量不超过 ZP。LG 首年预测未必通过观测点。Park 第二年实际 LG 为 35.95 点，校正预测 38.36 点，二次预测 63.18 点；同年 TCC 尚缺，不能验证 DE。

未来观测以缺测表示。阴影为训练留出误差形成的经验范围，不能对近期火灾作保证概率解释。

6.2 Trajectories for Park and Dixie

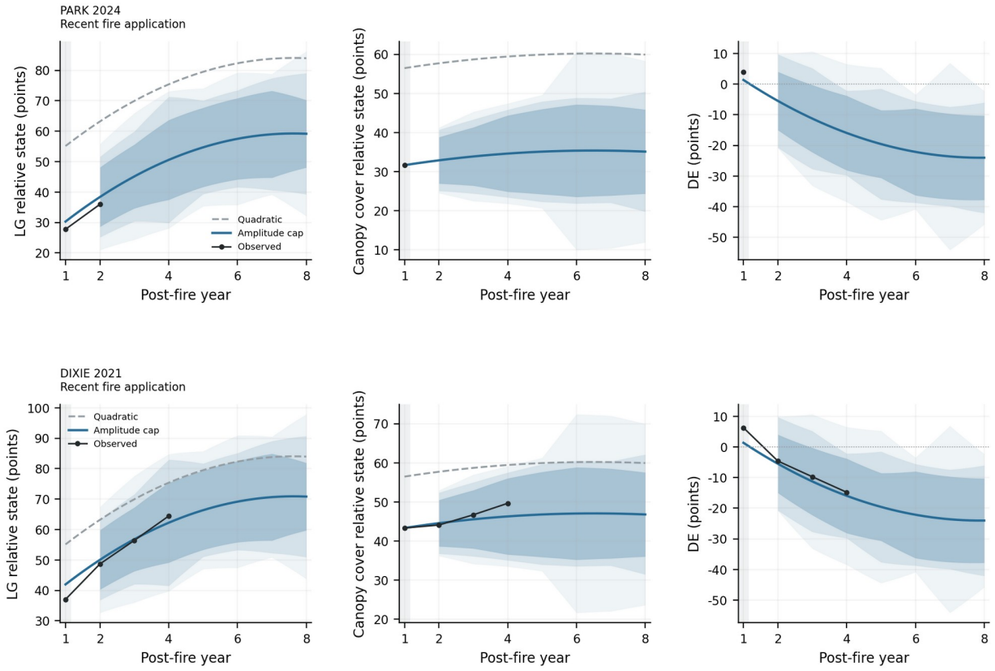


Figure 2. Dashed grey lines show the quadratic functions, blue lines the amplitude cap correction, and black points the actual remotely sensed observations.

Park and Dixie receive the full ZP_TCC shift. Their LG shifts retain the direction of the LG error and do not exceed the ZP_TCC shift in magnitude. The first-year LG prediction need not pass through the observation. Park's actual second-year LG is 35.95 points; the corrected prediction is 38.36 points and the quadratic prediction 63.18 points. TCC is still missing for that year, preventing validation of DE.

Future observations are left missing. Shading shows empirical ranges derived from held-out training errors; it is not a guarantee of the stated probability for these recent fires.

6.3 Caldor 与 Creek 的走势

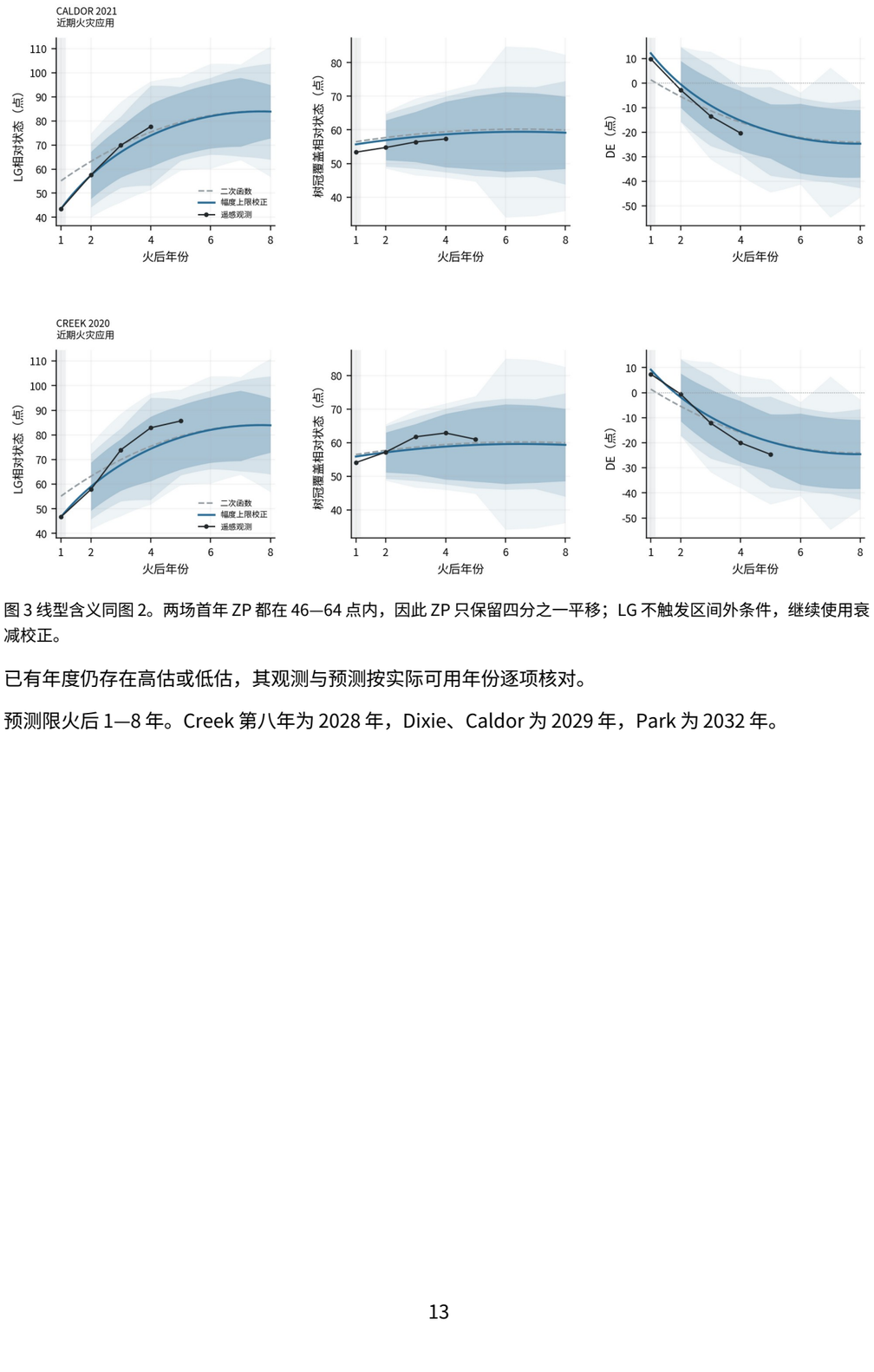


图 3 线型含义同图 2。两场首年 ZP 都在 46—64 点内，因此 ZP 只保留四分之一平移；LG 不触发区间外条件，继续使用衰减校正。

已有年度仍存在高估或低估，其观测与预测按实际可用年份逐项核对。

预测限火后 1—8 年。Creek 第八年为 2028 年，Dixie、Caldor 为 2029 年，Park 为 2032 年。

6.3 Trajectories for Caldor and Creek

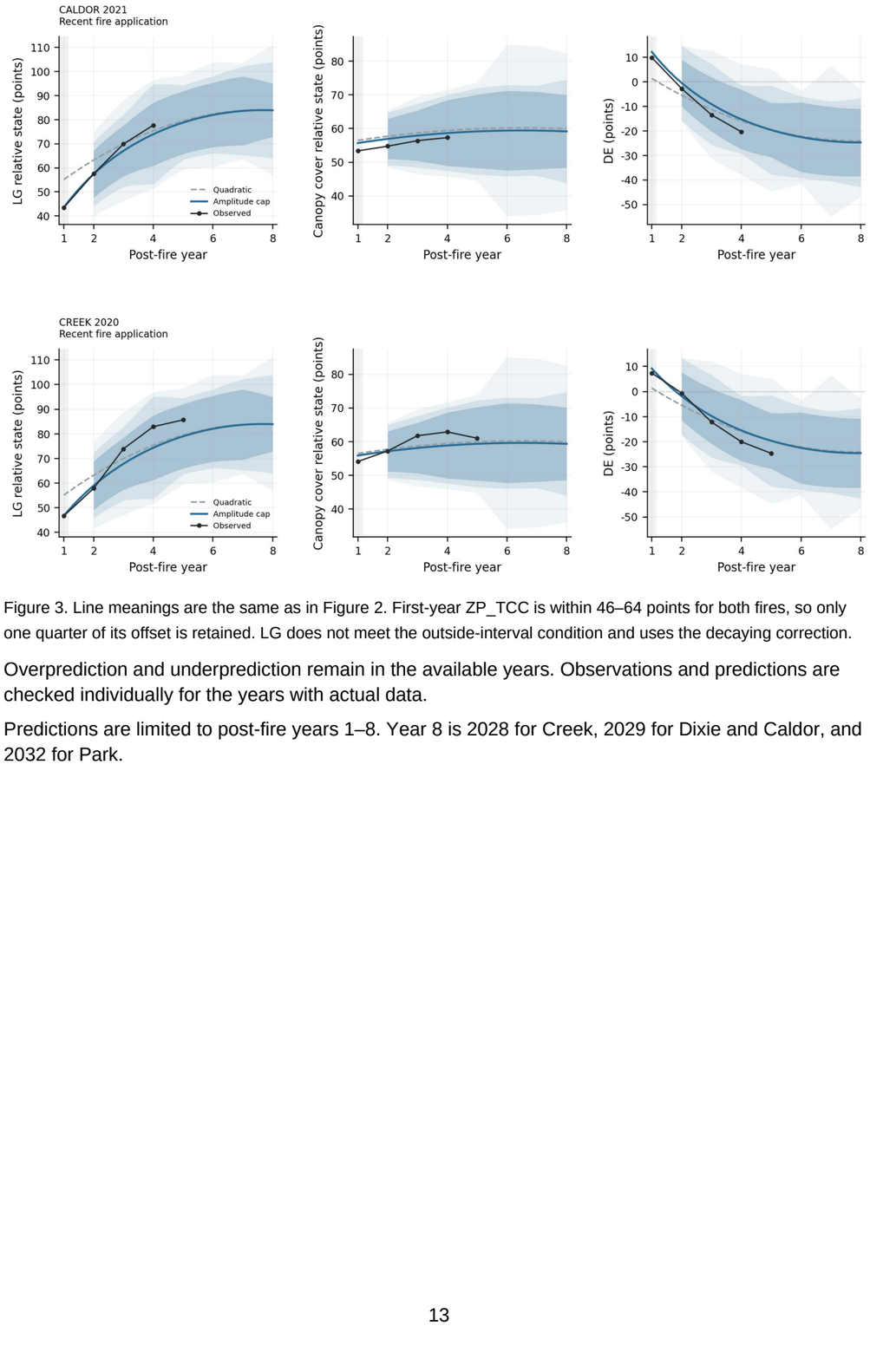


Figure 3. Line meanings are the same as in Figure 2. First-year ZP_TCC is within 46–64 points for both fires, so only one quarter of its offset is retained. LG does not meet the outside-interval condition and uses the decaying correction.

Overprediction and underprediction remain in the available years. Observations and predictions are checked individually for the years with actual data.

Predictions are limited to post-fire years 1–8. Year 8 is 2028 for Creek, 2029 for Dixie and Caldor, and 2032 for Park.

6.4 数值预测与适用边界

火灾 第八年日历年	LG 预测	ZP 预测	DE 预测
PARK 2032	59.14	35.11	-24.03
DIXIE 2029	70.81	46.78	-24.03
CALDOR 2029	83.87	59.16	-24.71
CREEK 2028	83.90	59.33	-24.56

表 8 火后第八年幅度上限校正预测，单位为点。DE 由两项相减，不另拟合；全部第 2—8 年预测和已有观测分列在资料表中。

近期案例未参与函数拟合。其已有第 2—8 年观测分别用于核对，结果与 37 场回顾性检验分开报告。

首年偏差的平移、上限与衰减都是待检验的数学假设，不能证明某种生态过程会以固定差距延续。缺少的实际 TCC 没有用预测值补齐。

6.4 Numerical predictions and limits of application

Fire and eighth-year calendar date	LG prediction	ZP prediction	DE prediction
PARK 2032	59.14	35.11	-24.03
DIXIE 2029	70.81	46.78	-24.03
CALDOR 2029	83.87	59.16	-24.71
CREEK 2028	83.90	59.33	-24.56

Table 8. Amplitude cap predictions for post-fire year 8, in points. DE is obtained by subtracting the two indicators, not by fitting another function. All predictions for years 2–8 and available observations are listed separately in the data tables.

The recent cases are not used to fit the functions. Their available observations from years 2–8 are checked individually, with results reported separately from the 37-fire retrospective test.

The shift, cap and decay of first-year offsets are mathematical hypotheses to be tested. They do not prove that an ecological process maintains a fixed difference. Missing actual TCC values are not filled with predictions.

7 测量证据与研究局限

7.1 理论结构目标的独立证据

NEON 逐树记录已能重算 32 个地点的 330 次活树断面积调查。共同 309 条、29 地点的水平参照中，多源模型有一定改善；56 组、21 地点的同址变化中，则未稳定优于简单方法。复测组不是 56 场独立火灾。[7—9]

因此本文采用 TCC 覆盖比例作为直接遥感代理，没有将航空高度或光谱量加权后冒称真实活立木断面积。LiDAR 可能包含站立枯木，树冠覆盖与活树数量、高度及断面积不能互换。

LG 固定参照只有 19 条、11 个空间组；131 条、21 地点的小样方航空迁移检验中，水平 RMSE 约 0.310，偏差约−0.284。尺度、季节及分类规则不同，部分差异无法归于单一原因。标准化不能保证消除随火烧和植被组成改变的偏差。

7.2 训练范围与不同变化轨迹

训练样本为 36 场 DE 呈下降轨迹的火灾，未覆盖全部可能的变化过程。检验样本中，DE 基本不变和增加事件仍有明显偏差，说明模型存在适用范围限制。

没有独立证据证明 DE 几乎不变或增加必然来自外部干扰。幅度上限校正在这批 37 场中的整体 DE 误差较低，但基本不变和增加事件仍存在较大偏差；不能把整体改善解释为适合所有轨迹。

7.3 验证与外推的限制

模型与校正规则来自回顾性研究资料，当前检验不具备严格前瞻盲测条件。36 场训练火灾分布于 10 个起火年份组，有限的事件数量和轨迹覆盖使预测能力评估仍有较大不确定性。

预测严格限定于火后 1—8 年。产品误差、空间相关、同一区域共同气候、对照变化及后续干扰均可能影响结果；遥感关联和函数形状不能单独证明生态因果机制。

7 Measurement evidence and study limitations

7.1 Independent evidence for the physical structural target

NEON individual-tree records allow 330 live-tree basal-area surveys from 32 sites to be recalculated. In the common set of 309 level references from 29 sites, a multisource model shows some improvement. For 56 paired changes at the same locations across 21 sites, it does not consistently outperform simple methods. Those repeat-measurement pairs are not 56 independent fires. [7–9]

TCC fraction is therefore used as a direct remotely sensed proxy. Airborne height and spectral measurements are not combined and presented as actual live-tree basal area. LiDAR may include standing dead trees; canopy cover, live-tree counts, height and basal area are not interchangeable.

The fixed LG calibration contains only 19 references in 11 spatial groups. In an airborne transfer check involving 131 small-plot records from 21 sites, RMSE for the measured level is approximately 0.310 and bias approximately −0.284. Spatial scale, season and classification rules differ, and some discrepancies cannot be attributed to a single cause. Normalization does not guarantee removal of bias that changes with fire effects or vegetation composition.

7.2 Training coverage and contrasting trajectories

The training sample consists of 36 fires with declining DE trajectories and does not cover all possible patterns of change. Test fires with approximately unchanged or increasing DE still show substantial bias, demonstrating limits to the model's applicability.

There is no independent evidence that approximately unchanged or increasing DE must result from external disturbance. The amplitude cap correction has lower overall DE error in these 37 test fires, but substantial errors remain for unchanged and increasing trajectories. Overall improvement should not be interpreted as suitability for every trajectory.

7.3 Limits of validation and extrapolation

The model and correction rules derive from retrospective study information; the current evaluation does not satisfy the conditions of a strictly prospective blind test. The 36 training fires span 10 ignition-year groups. The limited number of events and incomplete trajectory coverage leave considerable uncertainty in the assessment of predictive performance.

Predictions are strictly restricted to post-fire years 1–8. Product errors, spatial correlation, shared regional climate, changes in controls and subsequent disturbances may all influence results. Remotely sensed associations and function shapes alone cannot establish ecological causal mechanisms.

8 结论

LG 与树冠覆盖是不同的遥感指标。77 场具有完整八年记录的火灾表现出不同的平均时间轨迹，单场 DE 可下降、基本不变或增加，平均曲线不能代表所有事件。

37 场回顾性检验中，幅度上限校正的 LG、树冠覆盖和 DE RMSE 分别为 8.56、8.42 和 8.60 点，R^2 分别约 0.616、0.621 和 0.262，三项目标的误差均低于二次函数。训练验证中 DE 误差则略高于二次函数，尚不能确认普遍预测优势。

幅度上限校正利用第一年观测，在固定条件下限制 LG 的平移幅度，以保留偏差方向并避免两项校正量过度分离。其作用是调整遥感相对状态的数学预测，不能替代对实际森林结构与生态过程的独立测量。

近期四场火灾的应用展示了该方法的计算方式和观测限制。物理 ZP、树冠覆盖代理与绿色覆盖估计具有不同含义，研究结论限于这些观测指标在火后八年内的相对变化。

作者贡献与 AI 工具使用说明

作者负责提出研究问题与研究目标，确定指标和模型形式，提出首年校正及幅度上限思路，选择正式方法，并审定结果解释和研究结论。

ChatGPT 协助搜集公开数据和文献、处理与核对数据，以及编写数据获取、统计计算、作图和复现程序，并按作者要求整理研究资料。论文最终内容与结论由作者核查并负责。

8 Conclusions

LG and canopy cover are distinct remotely sensed indicators. The 77 fires with complete eight-year records show different mean temporal trajectories for the two indicators. Individual DE trajectories may decline, stay approximately unchanged or increase; the mean curve cannot represent every event.

Across the 37 retrospective test fires, amplitude cap RMSE is 8.56 points for LG, 8.42 for canopy cover and 8.60 for DE, with R^2 approximately 0.616, 0.621 and 0.262. All three errors are lower than for the quadratic functions. In training validation, however, DE error is slightly higher than for the quadratic function, so a general predictive advantage cannot yet be confirmed.

The amplitude cap correction uses first-year observations and limits the LG shift under fixed conditions, preserving its direction while avoiding excessive separation of the two corrections. It adjusts mathematical predictions of remotely sensed relative states and cannot replace independent measurements of actual forest structure or ecological processes.

Applications to four recent fires illustrate the calculation and the limits of available observations. Physical ZP, the canopy-cover proxy and estimated green cover have different meanings. The conclusions are restricted to relative changes in these observed indicators within eight years after fire.

Author contributions and use of AI tools

The author formulated the research questions and objectives, determined the indicators and model forms, proposed the first-year correction and amplitude cap approach, selected the adopted method, and reviewed the interpretation and research conclusions.

ChatGPT assisted with collecting public data and literature, processing and checking data, and writing programs for data acquisition, statistical calculation, plotting and reproduction. It also organized research materials at the author's request. The author reviews and takes responsibility for the final content and conclusions.

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[12] CAL FIRE. Dixie Fire, started 13 July 2021. 数据或原文

[13] CAL FIRE. Caldor Fire, started 14 August 2021. 数据或原文

[14] CAL FIRE. Creek Fire, started 4 September 2020. 数据或原文

以上近期事件资料与 TCC 年份覆盖于 2026 年 10 月 24 日复核。火灾日期核对不等于已取得其生态地面真值；数值预测来自所记录版本的固定位置遥感资料。

9 References and data sources

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[11] CAL FIRE. Park Fire, started 24 July 2024. Source

[12] CAL FIRE. Dixie Fire, started 13 July 2021. Source

[13] CAL FIRE. Caldor Fire, started 14 August 2021. Source

[14] CAL FIRE. Creek Fire, started 4 September 2020. Source

The recent-event information and TCC year coverage were checked on 24 October 2026. Verifying fire dates does not establish the availability of ecological ground truth. Numerical predictions use fixed-location remotely sensed data from the recorded product versions.

附录 A 指标与计算量的详细说明

A.1 NDVI EVI 与 LG

反射率 ρ 表示某波段反射辐射与入射辐射的比值。ρ_R 为红光反射率，ρ_NIR 为近红外反射率，ρ_B 为蓝光反射率。绿色叶片对红光和近红外的响应不同，NDVI 利用这种差异；EVI 增加蓝光和修正项，以减小部分大气及背景影响。它们都不是活树数量或生态恢复程度。

$$NDVI = \frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + \rho_R}$$

$$EVI = 2.5\frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + 6\rho_R - 7.5\rho_B + 1}$$

MOD13Q1 的 NDVI 和 EVI 原值乘 0.0001 转换到指数尺度。研究先筛选每年 6—8 月 SummaryQA 为 0 或 1 的观测，再分别取 NDVI 和 EVI 中位数；严格 QA 为 0 的次数另记录。多数位置每年有 6 次可用合成记录，最少 3 次。合成次数不等于实际无云天数。

LG 的固定计算式如下，N 和 E 分别表示上述 NDVI 和 EVI；clip 表示小于 0 的估计取 0、大于 1 取 1，保持绿色覆盖估计的有界定义。这个操作只作用于原始 LG 估计，不作用于相对状态 R 或 DE。

$$LG = \mathrm{clip}_{[0,1]}(-0.149100557 + 0.554620599N + 0.727351848E)$$

LG 两个系数来自 DIRECT 参照校准。19 条、11 个空间组参照的空间分组验证 RMSE 约 0.114，NDVI 单变量约 0.129；参照数量有限，航空迁移检验的明显偏差表明测量适用性仍有限。

A.2 TCC 与 ZP_TCC

TCC 为树冠覆盖百分比，ZP_TCC＝TCC／100 为 0—1 比例。例如 TCC＝60 表示覆盖百分比 60%，ZP_TCC＝0.60；都还不是标准化相对状态。使用 Science_Percent_Tree_Canopy_Cover 字段，data_mask＝1；不同于经进一步后处理的 NLCD 同名覆盖产品。

30 米有效覆盖按面积平均到 250 米网格，再在固定位置之间平均。少量网格只有部分 30 米面积有效，有效面积比例记录于质量数据表。

TCC 的产品误差字段不能直接解释为活树断面积误差。本研究不据此加权时间回归，也不把覆盖变化自动等同于某棵树死亡或新生。

Appendix A Definitions of indicators and calculation quantities

A.1 NDVI EVI and LG

Reflectance ρ is the ratio of reflected to incoming radiation in a spectral band. ρ_R denotes red reflectance, ρ_NIR near-infrared reflectance, and ρ_B blue reflectance. Green leaves respond differently in the red and near-infrared bands; NDVI uses this difference. EVI adds a blue band and correction terms to reduce some atmospheric and background effects. Neither index measures live-tree counts or the extent of ecological recovery.

$$NDVI = \frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + \rho_R}$$

$$EVI = 2.5\frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + 6\rho_R - 7.5\rho_B + 1}$$

Raw MOD13Q1 NDVI and EVI values are multiplied by 0.0001 to obtain index values. Records from June–August with SummaryQA of 0 or 1 are selected, and NDVI and EVI medians are calculated separately. Counts with the stricter QA value of 0 are also recorded. Most locations have six usable composite records per year, with a minimum of three. The number of composites is not the number of actual cloud-free days.

The fixed LG formula is shown below. N and E denote NDVI and EVI as defined above. The clip operation sets estimates below 0 to 0 and above 1 to 1, preserving the bounded definition of estimated green cover. It acts only on raw LG, not on the relative state R or DE.

$$LG = \mathrm{clip}_{[0,1]}(-0.149100557 + 0.554620599N + 0.727351848E)$$

The two LG coefficients are calibrated against DIRECT references. Spatially grouped validation using 19 records in 11 spatial groups gives RMSE of approximately 0.114, compared with approximately 0.129 for NDVI alone. The reference sample is small, and the clear bias in airborne transfer testing indicates limited measurement applicability.

A.2 TCC and ZP_TCC

TCC is percentage tree canopy cover; ZP_TCC = TCC/100 is the corresponding fraction between 0 and 1. For example, TCC = 60 means 60% canopy cover and ZP_TCC = 0.60. Neither is yet a normalized relative state. The Science_Percent_Tree_Canopy_Cover field with data_mask = 1 is used. This differs from the similarly named NLCD cover product, which undergoes further post-processing.

Valid 30 m cover values are area-averaged onto the 250 m grid, then averaged across fixed locations. Some grid cells have valid values over only part of their 30 m area; valid-area fractions are retained in the quality records.

The TCC product-error field cannot be interpreted directly as error in live-tree basal area. It is not used to weight the time regression, and a change in cover is not automatically interpreted as the death or establishment of an individual tree.

A.3 物理 ZP 与地面调查

$$ZP_{\text{phys}} = \frac{10000}{A} \sum_{i=1}^{n} \pi\left(\frac{DBH_i}{200}\right)^2 \quad (\text{m}^2/\text{ha})$$

A 为实际调查面积，单位平方米；DBH_i 为第 i 棵符合条件活树的胸径，单位厘米；n 为调查范围内符合条件的活树数。每棵树断面积为 π(DBH_i／200)² 平方米，求和后乘 10,000／A 换算为平方米／公顷。多干树按调查协议中可辨认的树干处理，不将失踪或状态未知当成死亡。

胸径通常在地面以上 130 厘米处测量，特殊树形按协议在记录位置测径。复测位置变化会影响直径差，不能把全部差异直接当作生长。物理 ZP 可通过逐树调查直接计算；限制是其地面覆盖和全数据库遥感反演尚不足，不是“无法测量”。

A.4 相对状态的数值含义

I_k(t)、C_k(t)分别为指标 k 在固定受灾和对照位置上的当年平均。q_k＝I_k／C_k；b_k 为灾前三年 q 的算术平均；R_k＝q_k／b_k。t＝1 表示起火后的下一个日历年，负年份用于基准，t＝0 不进入主时间预测。

例如，某项指标灾前三年受灾／对照比值平均为 0.80，火后某年比值为 0.64，则 R＝0.64／0.80＝0.80，展示为 80 点。它不等于原始覆盖率 80%，也不等于整个生态系统恢复 80%。若对照也下降，相对状态可能比只看受灾原值更高。

DE＝R_ZP_TCC－R_LG，乘 100 后以点计。两项分别标准化，所以两者相等并不表示原始覆盖值相等。DE 也不包含时间导数，不能直接称为恢复速度差。

基准变异系数为灾前三年 q 的样本标准差除以其平均绝对值，用于检查参考是否稳定。低基准 0.05 及变异系数 20%在计算前设为质量提示，不作为删除异常火灾的理由。

A.5 误差和不确定范围

每场火灾先计算各年度平方误差、绝对误差和有符号误差的平均，再对火灾平均。RMSE 为平均平方误差的平方根，MAE 为平均绝对误差，偏差为平均有符号误差。观测标准差使用相同火灾等权规则，与 RMSE 保持同一尺度。

重抽样每次抽取一组起火年份，同一年份的火灾一起进入；重复 10,000 次，取结果的 2.5%和 97.5%分位数作为 95%范围。这种范围表达样本不确定性，不证明所有火灾彼此独立，也不消除产品系统偏差。正文案例图的范围为单场预测的条件经验范围，不能解释为总体平均值的置信范围。

A.3 Physical ZP and field inventories

$$ZP_{\text{phys}} = \frac{10000}{A} \sum_{i=1}^{n} \pi\left(\frac{DBH_i}{200}\right)^2 \quad (\text{m}^2/\text{ha})$$

A is the actual surveyed area in square metres; DBH_i is the diameter at breast height, in centimetres, of the ith eligible living tree; n is the number of eligible living trees in the surveyed area. Each tree's basal area is π(DBH_i/200)² square metres. Multiplying the sum by 10,000/A converts it to square metres per hectare. Identifiable stems of multistem trees are handled according to the inventory protocol. Missing trees or unknown status are not treated as deaths.

Diameter at breast height is normally measured 130 cm above ground. For unusual tree forms, the protocol specifies a recorded measurement position. Changes in that position between surveys can affect diameter differences, so not every difference can be interpreted as growth. Physical ZP can be calculated directly from individual-tree inventories; the limitation is inadequate field coverage and remote-sensing retrieval across the full database, not an inability to measure it.

A.4 Numerical meaning of relative states

I_k(t) and C_k(t) are the annual means of indicator k at fixed burned and control locations. q_k = I_k/C_k; b_k is the arithmetic mean of q over the three pre-fire years; R_k = q_k/b_k. Here t = 1 means the calendar year after the ignition year. Negative years define the baseline; t = 0 is excluded from the main time predictions.

For example, if an indicator's mean pre-fire burned-to-control ratio is 0.80 and its ratio in a post-fire year is 0.64, then R = 0.64/0.80 = 0.80, displayed as 80 points. This is neither raw cover of 80% nor 80% recovery of the whole ecosystem. If the control also declines, the relative state may be higher than suggested by the burned-area value alone.

DE = R_ZP_TCC − R_LG, multiplied by 100 for display in points. Because the two indicators are normalized separately, equal relative states do not imply equal raw cover values. DE does not contain a time derivative and cannot directly be called a difference in recovery rates.

The baseline coefficient of variation is the sample standard deviation of q over the three pre-fire years divided by the absolute value of its mean. It checks reference stability. A low-baseline threshold of 0.05 and coefficient-of-variation threshold of 20% were set as quality flags before calculation, not as reasons to delete unusual fires.

A.5 Errors and uncertainty ranges

Squared, absolute and signed errors are first averaged across the available years within each fire and then across fires. RMSE is the square root of mean squared error; MAE is mean absolute error; bias is mean signed error. Observed SD uses the same equal-fire weighting, keeping it on the same scale as RMSE.

Resampling draws ignition-year groups, with fires from the same year entering together. The procedure is repeated 10,000 times; the 2.5% and 97.5% quantiles define a 95% range. This describes sample uncertainty, does not establish that every fire is independent, and does not remove systematic product bias. Ranges in the main case figures are conditional empirical ranges for individual predictions, not confidence intervals for the population mean.

附录 B 训练与主评分事件名单

训练火灾	年份	检验火灾	年份
POWER	2004	NUTALL COMPLEX (NUTALL)	2004
I-90 COMPLEX (TARKIO)	2005	WILLOW	2004
VALLEY ROAD	2005	BLOSSOM	2005
CAVITY LAKE	2006	SIGNAL ROCK	2005
HEAVENS GATE COMPLEX (BLACK BUTTE)	2006	COLUMBIA COMPLEX	2006
SOUTHFORK COMPLEX (COUGAR)	2006	JUNGLE	2006
THORN CREEK	2006	RED EAGLE	2006
CORPORAL	2007	TATOOSH COMPLEX	2006
MOONLIGHT	2007	BRUSH CREEK	2007
RAILLEY MOUNTAIN	2007	CASTLE ROCK	2007
ROMBO MOUNTAIN	2007	CHIPPY CREEK	2007
SHOWER BATH COMPLEX (SHOWER BATH)	2007	CONFLUENCE COMPLEX (CLEAR SAGE)	2007
SLIDE	2007	ELK COMPLEX (KING CREEK 2)	2007
CANYON COMPLEX (SCOTCH)	2008	GRAYS CREEK	2007
IRON AND ALPS COMPLEXES (EAGLE)	2008	JOCKO LAKES	2007
LIME COMPLEX (LIME)	2008	MANILA CREEK	2007
LIME COMPLEX (MINERS)	2008	CHALK	2008
SHU LIGHTNING COMPLEX (MOON)	2008	CORRAL	2008

36 场用于拟合与训练交叉验证，37 场用于回顾性检验。8 场补充记录及全部 81 场的用途见 data/events.csv。

Appendix B Training and main test fires

Training fire	Year	Test fire	Year
POWER	2004	NUTALL COMPLEX (NUTALL)	2004
I-90 COMPLEX (TARKIO)	2005	WILLOW	2004
VALLEY ROAD	2005	BLOSSOM	2005
CAVITY LAKE	2006	SIGNAL ROCK	2005
HEAVENS GATE COMPLEX (BLACK BUTTE)	2006	COLUMBIA COMPLEX	2006
SOUTHFORK COMPLEX (COUGAR)	2006	JUNGLE	2006
THORN CREEK	2006	RED EAGLE	2006
CORPORAL	2007	TATOOSH COMPLEX	2006
MOONLIGHT	2007	BRUSH CREEK	2007
RAILLEY MOUNTAIN	2007	CASTLE ROCK	2007
ROMBO MOUNTAIN	2007	CHIPPY CREEK	2007
SHOWER BATH COMPLEX (SHOWER BATH)	2007	CONFLUENCE COMPLEX (CLEAR SAGE)	2007
SLIDE	2007	ELK COMPLEX (KING CREEK 2)	2007
CANYON COMPLEX (SCOTCH)	2008	GRAYS CREEK	2007
IRON AND ALPS COMPLEXES (EAGLE)	2008	JOCKO LAKES	2007
LIME COMPLEX (LIME)	2008	MANILA CREEK	2007
LIME COMPLEX (MINERS)	2008	CHALK	2008
SHU LIGHTNING COMPLEX (MOON)	2008	CORRAL	2008

Of the fires listed, 36 are used for fitting and training cross-validation, and 37 for retrospective testing. The eight supplementary records and the uses of all 81 fires are recorded in data/events.csv.

附录 B 训练与检验火灾名单 续

训练火灾	年份	检验火灾	年份
FREEZE OUT	2012	IRON AND ALPS COMPLEXES (IRONSIDE)	2008
MILLIE	2012	RATTLE	2008
MUSTANG COMPLEX	2012	YOLLA BOLLY COMPLEX (IRON)	2008
READING	2012	BRUSHY MOUNTAIN	2009
WHITEWATER-BALDY	2012	HIGHWAY 31	2009
AMERICAN	2013	LA BREA	2009
MILE MARKER 28	2013	NORTH FORK COMPLEX	2009
RIM	2013	SCHULTZ	2010
WHISKEY	2013	CHIPS	2012
KING	2014	NORTH PASS	2012
MAN	2014	DAY	2014
ROUGH	2015	HAPPY CAMP COMPLEX	2014
SLIDE	2015	WHISKEY COMPLEX	2014
TRAIL CREEK	2015	ELDORADO	2015
WOLVERINE	2015	TOWER	2015
DEEP CREEK	2017	COPPER KING	2016
HELENA	2017	BRIANHEAD	2017
JULY	2017	HANOVER	2017
—	—	SCALP	2017

训练交叉验证按起火年份分组；回顾性检验使用 36 场全训练样本确定的二次函数和相同校正规则。

Appendix B Training and test fires continued

Training fire	Year	Test fire	Year
FREEZE OUT	2012	IRON AND ALPS COMPLEXES (IRONSIDE)	2008
MILLIE	2012	RATTLE	2008
MUSTANG COMPLEX	2012	YOLLA BOLLY COMPLEX (IRON)	2008
READING	2012	BRUSHY MOUNTAIN	2009
WHITEWATER-BALDY	2012	HIGHWAY 31	2009
AMERICAN	2013	LA BREA	2009
MILE MARKER 28	2013	NORTH FORK COMPLEX	2009
RIM	2013	SCHULTZ	2010
WHISKEY	2013	CHIPS	2012
KING	2014	NORTH PASS	2012
MAN	2014	DAY	2014
ROUGH	2015	HAPPY CAMP COMPLEX	2014
SLIDE	2015	WHISKEY COMPLEX	2014
TRAIL CREEK	2015	ELDORADO	2015
WOLVERINE	2015	TOWER	2015
DEEP CREEK	2017	COPPER KING	2016
HELENA	2017	BRIANHEAD	2017
JULY	2017	HANOVER	2017
—	—	SCALP	2017

Training cross-validation is grouped by ignition year. Retrospective testing uses the quadratic functions fitted to all 36 training fires and the same correction rules.

附录 C DE 的经验预测范围

火后年份	80%目标	90%目标	95%目标
2	±9.54／81.1%	±15.26／100.0%	±15.45／100.0%
3	±11.01／83.8%	±16.49／97.3%	±21.95／100.0%
4	±12.18／89.2%	±13.83／91.9%	±22.52／100.0%
5	±11.07／73.0%	±18.09／97.3%	±24.87／100.0%
6	±14.17／89.2%	±16.59／94.6%	±18.74／100.0%
7	±13.98／86.5%	±16.23／91.9%	±30.56／100.0%
8	±13.78／83.8%	±18.04／94.6%	±21.78／97.3%

每格为固定半宽（点）／37 场检验的实际覆盖率。同一指标与年份对所有案例使用相同半宽。

这些是累计目标，不是保证概率。第一年不画范围，区间不截断到 0—100 点。宽度从训练固定规则的留出残差计算，未用检验残差调整。

Appendix C Empirical prediction ranges for DE

Post-fire year	80% target	90% target	95% target
2	±9.54 81.1%	±15.26 100.0%	±15.45 100.0%
3	±11.01 83.8%	±16.49 97.3%	±21.95 100.0%
4	±12.18 89.2%	±13.83 91.9%	±22.52 100.0%
5	±11.07 73.0%	±18.09 97.3%	±24.87 100.0%
6	±14.17 89.2%	±16.59 94.6%	±18.74 100.0%
7	±13.98 86.5%	±16.23 91.9%	±30.56 100.0%
8	±13.78 83.8%	±18.04 94.6%	±21.78 97.3%

Each cell gives the fixed half-width in points / actual coverage in the 37 test fires. Every case uses the same half-width for a given indicator and year.

These are cumulative coverage targets, not guaranteed probabilities. No range is drawn for the first year, and intervals are not truncated to 0–100 points. Widths come from training held-out errors under the fixed rules and are not adjusted using test residuals.

附录 C LG 的经验预测范围

火后年份	80%目标	90%目标	95%目标
2	±9.78／83.8%	±13.26／86.5%	±17.42／97.3%
3	±10.50／75.7%	±14.80／94.6%	±20.82／100.0%
4	±13.12／78.4%	±20.76／97.3%	±22.54／97.3%
5	±13.02／89.2%	±15.36／97.3%	±19.37／97.3%
6	±13.35／86.5%	±15.95／97.3%	±21.79／100.0%
7	±14.28／91.9%	±18.42／100.0%	±19.85／100.0%
8	±11.10／75.7%	±19.91／100.0%	±27.09／100.0%

每格为固定半宽（点）／37 场检验的实际覆盖率。同一指标与年份对所有案例使用相同半宽。

这些是累计目标，不是保证概率。第一年不画范围，区间不截断到 0—100 点。宽度从训练固定规则的留出残差计算，未用检验残差调整。

Appendix C Empirical prediction ranges for LG

Post-fire year	80% target	90% target	95% target
2	±9.78 83.8%	±13.26 86.5%	±17.42 97.3%
3	±10.50 75.7%	±14.80 94.6%	±20.82 100.0%
4	±13.12 78.4%	±20.76 97.3%	±22.54 97.3%
5	±13.02 89.2%	±15.36 97.3%	±19.37 97.3%
6	±13.35 86.5%	±15.95 97.3%	±21.79 100.0%
7	±14.28 91.9%	±18.42 100.0%	±19.85 100.0%
8	±11.10 75.7%	±19.91 100.0%	±27.09 100.0%

Each cell gives the fixed half-width in points / actual coverage in the 37 test fires. Every case uses the same half-width for a given indicator and year.

These are cumulative coverage targets, not guaranteed probabilities. No range is drawn for the first year, and intervals are not truncated to 0–100 points. Widths come from training held-out errors under the fixed rules and are not adjusted using test residuals.

附录 C 树冠覆盖的经验预测范围

火后年份	80%目标	90%目标	95%目标
2	±5.94／83.8%	±7.82／94.6%	±8.41／97.3%
3	±7.47／81.1%	±9.47／91.9%	±11.43／91.9%
4	±9.76／81.1%	±11.23／83.8%	±12.84／89.2%
5	±10.89／81.1%	±12.85／89.2%	±14.51／89.2%
6	±11.84／83.8%	±13.52／89.2%	±25.41／100.0%
7	±11.50／81.1%	±13.34／89.2%	±25.01／94.6%
8	±10.77／78.4%	±15.35／89.2%	±23.20／91.9%

每格为固定半宽（点）／37 场检验的实际覆盖率。同一指标与年份对所有案例使用相同半宽。

这些是累计目标，不是保证概率。第一年不画范围，区间不截断到 0—100 点。宽度从训练固定规则的留出残差计算，未用检验残差调整。

Appendix C Empirical prediction ranges for canopy cover

Post-fire year	80% target	90% target	95% target
2	±5.94 83.8%	±7.82 94.6%	±8.41 97.3%
3	±7.47 81.1%	±9.47 91.9%	±11.43 91.9%
4	±9.76 81.1%	±11.23 83.8%	±12.84 89.2%
5	±10.89 81.1%	±12.85 89.2%	±14.51 89.2%
6	±11.84 83.8%	±13.52 89.2%	±25.41 100.0%
7	±11.50 81.1%	±13.34 89.2%	±25.01 94.6%
8	±10.77 78.4%	±15.35 89.2%	±23.20 91.9%

Each cell gives the fixed half-width in points / actual coverage in the 37 test fires. Every case uses the same half-width for a given indicator and year.

These are cumulative coverage targets, not guaranteed probabilities. No range is drawn for the first year, and intervals are not truncated to 0–100 points. Widths come from training held-out errors under the fixed rules and are not adjusted using test residuals.

附录 D R^2 的计算与解释

对同一组预测误差，先在各场火灾内部平均平方误差，再在火灾间等权，得到 MSE。RMSE 为其平方根。以同样权重计算观测平均值和方差，SD 为方差平方根。

$$R^2_{\text{test}} = 1 - \frac{\text{RMSE}^2}{\text{SD}^2}$$

本研究采用总体描述方差，不作 N−1 修正，因此上述比值与同权重的 1−残差平方和／总平方和相等。改变权重、时间窗口或样本后，不能混用原 RMSE 或 SD。

总体第 2—8 年的 R^2 与逐年的 R^2 分开报告。总体 SD 包括年度变化和火灾间差异；逐年 SD 只描述该年火灾间差异。负 R^2 保留，它表示比使用该批实际数据的平均值有更大平方误差。

R^2 不表示预测命中概率，不证明生态因果机制，也不代表物理 ZP 测量误差。小样本或 SD 很小的组会使比值不稳定，因此同时保留 RMSE、MAE、偏差、SD 和实际样本数。

Appendix D Calculating and interpreting R^2

For the same prediction errors, squared errors are first averaged within each fire and then equally across fires to obtain MSE. RMSE is its square root. The observed mean and variance are calculated using the same weights, and SD is the square root of that variance.

$$R^2_{\text{test}} = 1 - \frac{\text{RMSE}^2}{\text{SD}^2}$$

This study uses descriptive population variance without an N−1 adjustment. The displayed ratio therefore equals 1 minus the weighted residual sum of squares divided by the weighted total sum of squares. RMSE or SD values cannot be mixed across different weights, time windows or samples.

Overall R^2 for years 2–8 and annual R^2 are reported separately. Overall SD includes temporal change and differences among fires; annual SD describes only differences among fires in that year. Negative R^2 values are retained: they indicate larger squared error than using the mean of the actual observations in that sample.

R^2 is not a probability of prediction success, does not establish ecological causal mechanisms, and is not an error measure for physical ZP. Small samples or groups with very low SD can make the ratio unstable, so RMSE, MAE, bias, SD and actual sample sizes are also reported.

附录 E 可复现资料

复现程序位于 analysis/forest_recovery。data 目录保存固定位置、逐位置年度卫星值、事件用途、近期观测及测量参照资料；model.json 保存指标系数、二次函数系数和校正参数。

rebuild_measurements.py 计算 LG、树冠覆盖、受灾与对照均值及相对状态；analyze.py 计算二次函数和幅度上限校正的预测、误差和经验区间。make_figures.py 生成图表，build_report.py 与 build_manual.py 整理报告和技术说明。

predict.py 根据 LG 与 ZP_TCC 首年相对状态输出第 2—8 年的预测与经验范围。verify.py 检查指标、函数、图表和误差计算。完整程序与数据摘录见 forest_v5_code_and_evidence.pdf。

results/model_metrics.csv 和 annual_metrics.csv 记录 RMSE、MAE、偏差、SD 与 R^2；interval_half_widths.csv 和 test_interval_coverage.csv 分别记录半宽与实际覆盖。各文件均标明对应样本与预测方法。

all_case_predictions.csv 保存 81 场火灾第 1—8 年预测及已有观测；recent_predictions.csv 保存近期四场的应用。plot_values.csv 与图中数值对应。缺测观测保留为空值。

资料包包含可执行程序、数据字典和逐文件校验值。数据图证由实际 CSV 记录生成并注明来源，用于复核计算，不等同于独立地面观测。

Appendix E Reproducibility materials

Reproduction programs are in analysis/forest_recovery. The data directory contains fixed locations, annual satellite values for each location, event uses, recent observations and measurement references. model.json stores indicator coefficients, quadratic-function coefficients and correction parameters.

rebuild_measurements.py calculates LG, canopy cover, burned and control means, and relative states. analyze.py calculates predictions, errors and empirical intervals for the quadratic functions and amplitude cap correction. make_figures.py generates figures; build_report.py and build_manual.py assemble the report and technical manual.

predict.py uses first-year relative states of LG and ZP_TCC to output predictions and empirical ranges for years 2–8. verify.py checks the indicators, functions, plots and error calculations. Full programs and data excerpts are provided in forest_v5_code_and_evidence.pdf.

results/model_metrics.csv and annual_metrics.csv record RMSE, MAE, bias, SD and R^2. interval_half_widths.csv and test_interval_coverage.csv record half-widths and actual coverage, respectively. Each file identifies the corresponding sample and prediction method.

all_case_predictions.csv stores predictions for years 1–8 and available observations for all 81 fires. recent_predictions.csv records applications to the four recent fires. plot_values.csv corresponds to the plotted values. Missing observations remain empty.

The archive contains executable programs, a data dictionary and file checksums. Data-evidence figures are generated from actual CSV records with their sources identified. They support checking the calculations and are not independent field observations.